DO380

Exam v4.18

1. Troubleshooting POD Scheduling.

- Your work colleague accidentally misconfigured your OpenShift cluster prior to leaving on vacation. As a result, application pods are not able to run. Diagnose and correct the problem so that application pods are able to run on any worker node.
- You may use the image `quay.io/redhattraining/hello-world-nginx:v1.0` to verify your work.
- The image must deploy into a Running container without intervention.

Ans:-

```
Login into the cluster given admin user on Instruction.
------------------------------------------------------------>
$ oc login -u admin -p redhatocp --https://api.ocp4.example.com:6443
$ oc get pods -A | grep -iv running | grep -v completed
$ oc get nodes

A taint is applied to all the worker nodes. Modify the taint. (remove)
------------------------------------------------------------>
$ oc get nodes worker01 -oyaml | grep -i taint -A 4
taints:
- effect: NoSchedule
    key: key1
    value: value1

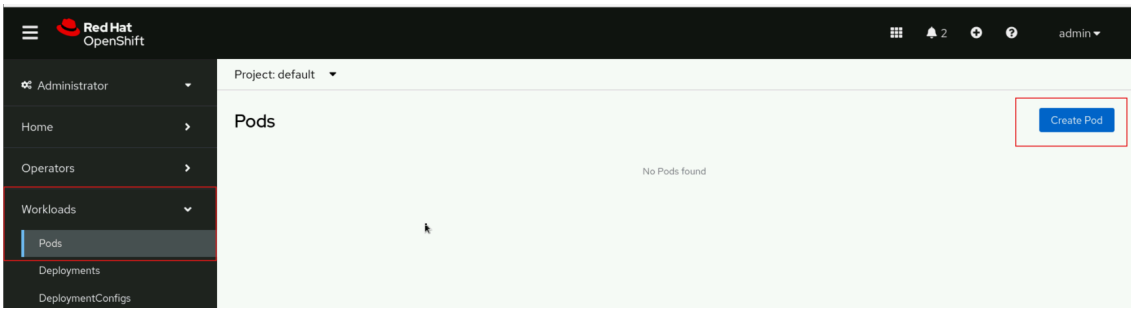
$ oc adm taint node worker01 key1-
$ oc adm taint node worker02 key1-
$ oc adm taint node worker03 key1-

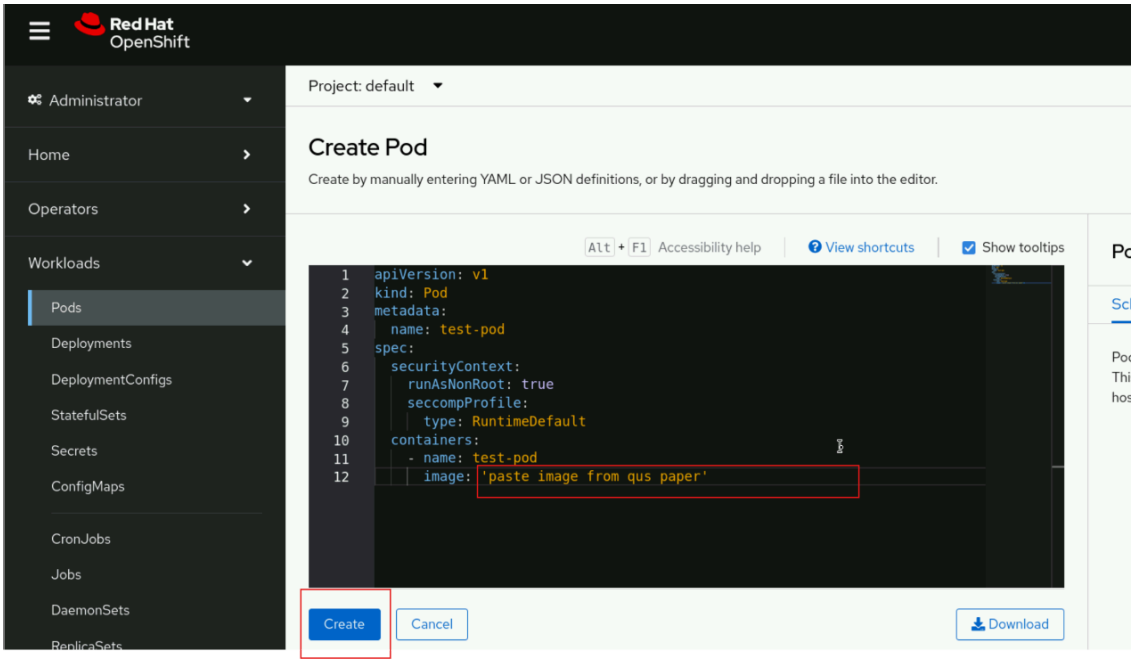
After removeing taint you should validate by executing below command.
------------------------------------------------------------------------>
$ oc get nodes worker01 -oyaml | grep -i taint -A 4
$ oc get nodes worker02 -oyaml | grep -i taint -A 4
$ oc get nodes worker03 -oyaml | grep -i taint -A

$ oc whoami --show-console
Login Into GUI deploy a pod as given and then validate.

$ oc get po -owide
```

```
Pod should be in running state in any of the worker node.
```





2. **Configure LDAP authentication**

- The LDAP server can be accessed with the following parameters.
  - Server Hostname `rhds.ocp4.example.com`
  - Bind DN `cn=Directory Manager`
  - Bind password `redhatocp`
  - Base DN `dc=example,dc=com`
  - Attributes `id=dn, email=mail, name=cn, prefferedusername=uid`
- Configure your OpenShift cluster with LDAP authentication using the following requirements.
  - Use default values for any other parameters you may need to configure.
  - The connection to the LDAP server must use TLS with the certificate authority file from `http://auth.domain/example.com/ca.crt`

- The user **kristendelgado** must be able to log in to the cluster using the password redhat123.

Ans:-

Note: - For pratice this qus in lab run 1.4. Guided Exercise

run (`lab start auth-ldap)`

- If you are doing by GUI and it's say to create configmap and secret with some name then remember to mentioned that name of secret and configmap to mentioned on oauth configuration.

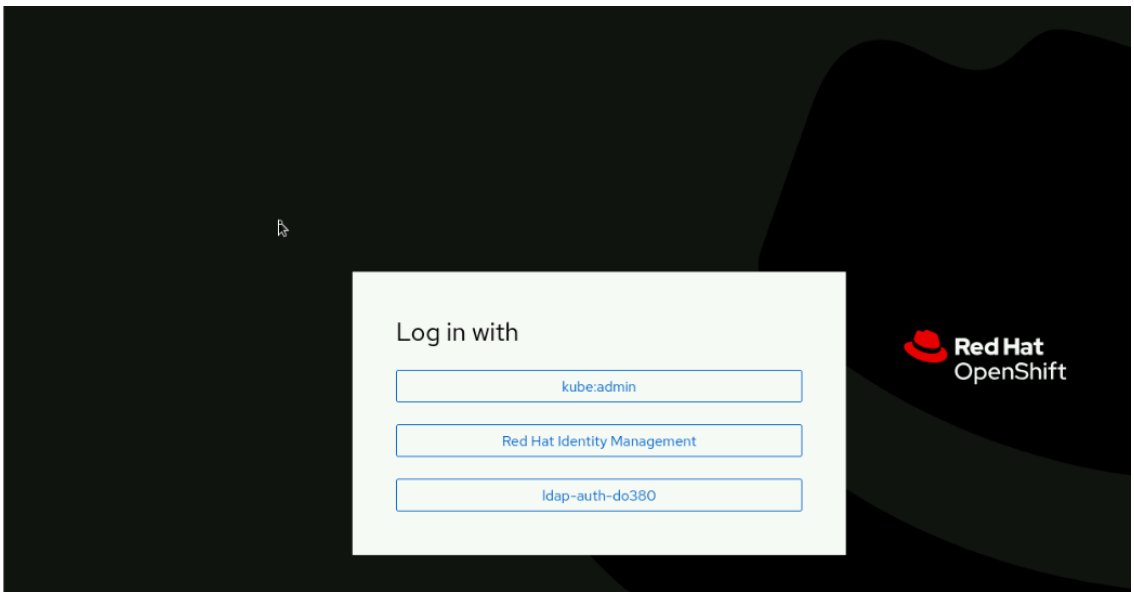
- In lab use ca.crt path as /home/student/DO380/labs/auth-ldap/rhds_ca.crt

```
$ curl -o ca.crt http://auth.domain/example.com/ca.crt(in exam download IT )
$ ls -lrth
  ca.crt
$ cat ca.crt
$ oc create cm ldap-ca --from-file=ca.crt=/home/student/DO380/labs/auth-ldap/
rhds_ca.crt -n openshift-config
$ oc -n openshift-config get cm
  NAME                                DATA   AGE
  ldap-ca                             1      13s
$ $ oc create secret generic ldap-secret --from-literal bindPassword=redhatoc
p -n openshift-config
$ oc -n openshift-config get secret
  NAME                                      TYPE
DATA   AGE
  ldap-secret                               Opaque
1      154s
$ oc get oauth cluster -oyaml > oauth.yaml
$ oc edit oauth cluster
  oauth.config.openshift.io/cluster edited

  spec:
      identityProviders:
    - ldap:
      attributes:
        email:
        - mail
        id:
        - dn
        name:
        - cn
        preferredUsername:
        - uid
      bindDN: Red Hat Directory Server
      bindPassword:
```

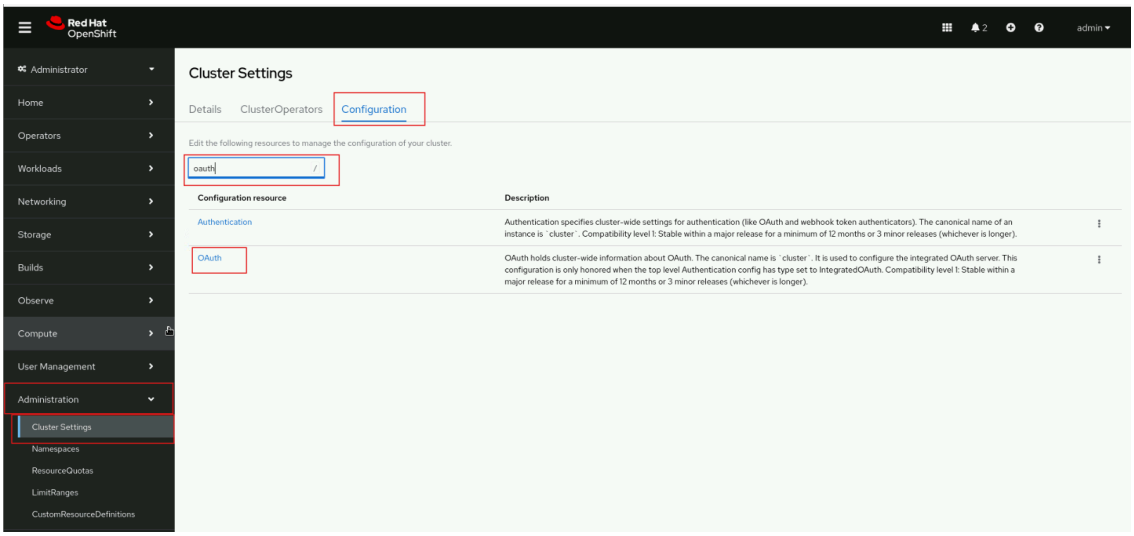
```
        name: ldap-secret
      ca:
        name: ldap-ca
      insecure: false
      url: ldap://rhds.ocp4.example.com/dc=example,dc=com?uid
    mappingMethod: claim
    name: ldap-auth-do380
    type: EOF

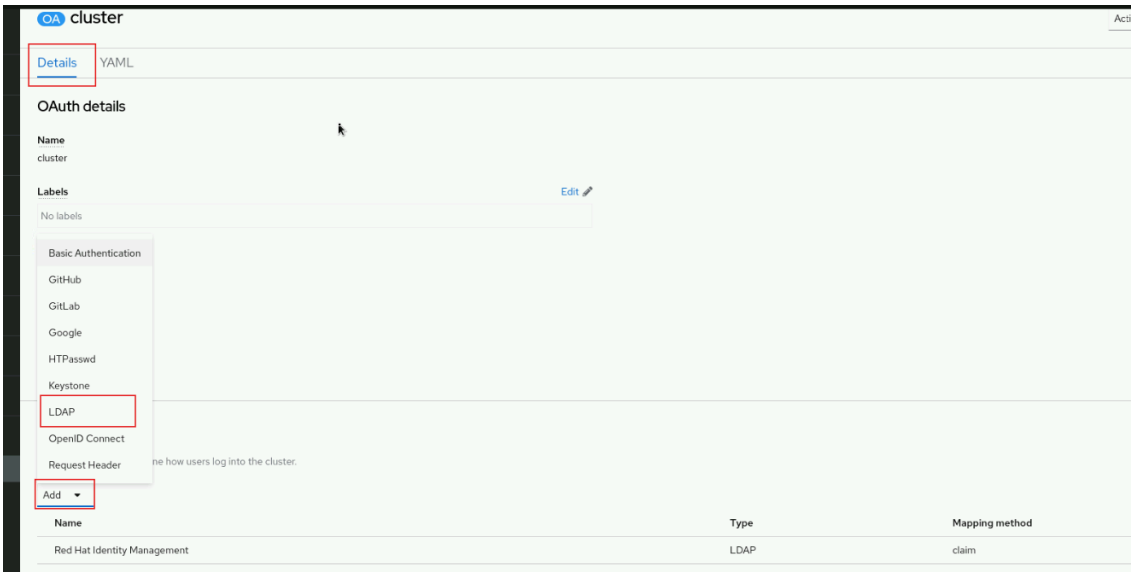
$ watch oc get po -n openshift-authentication
  pods will be restarted
$ oc whoami --show-console
  Check from GUI and login.
```

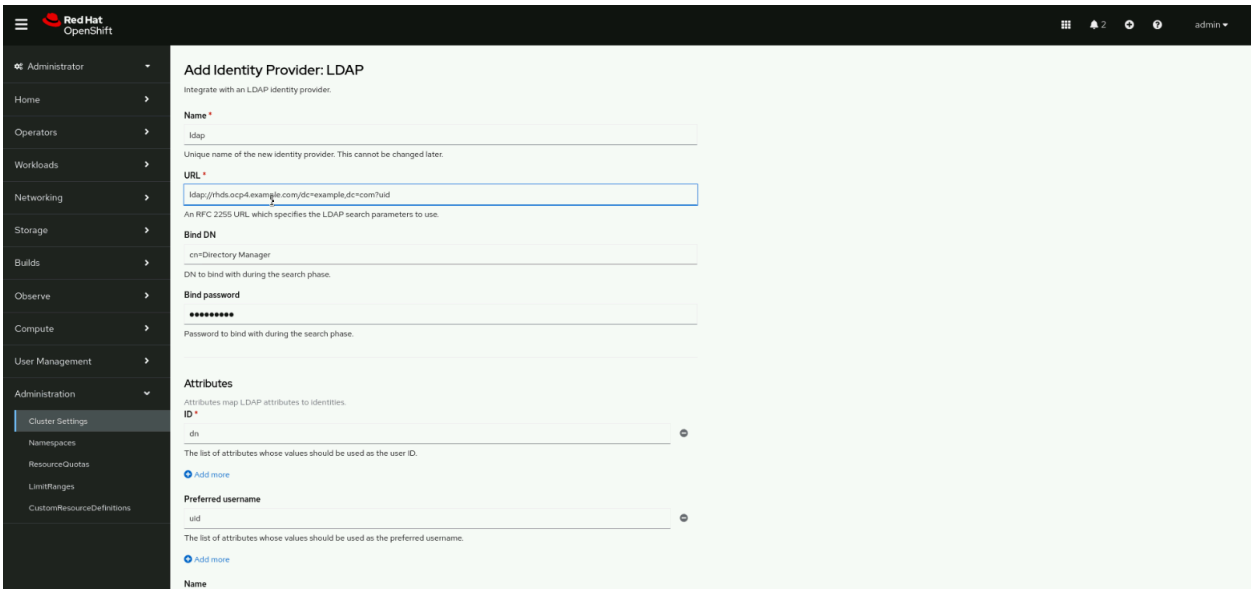


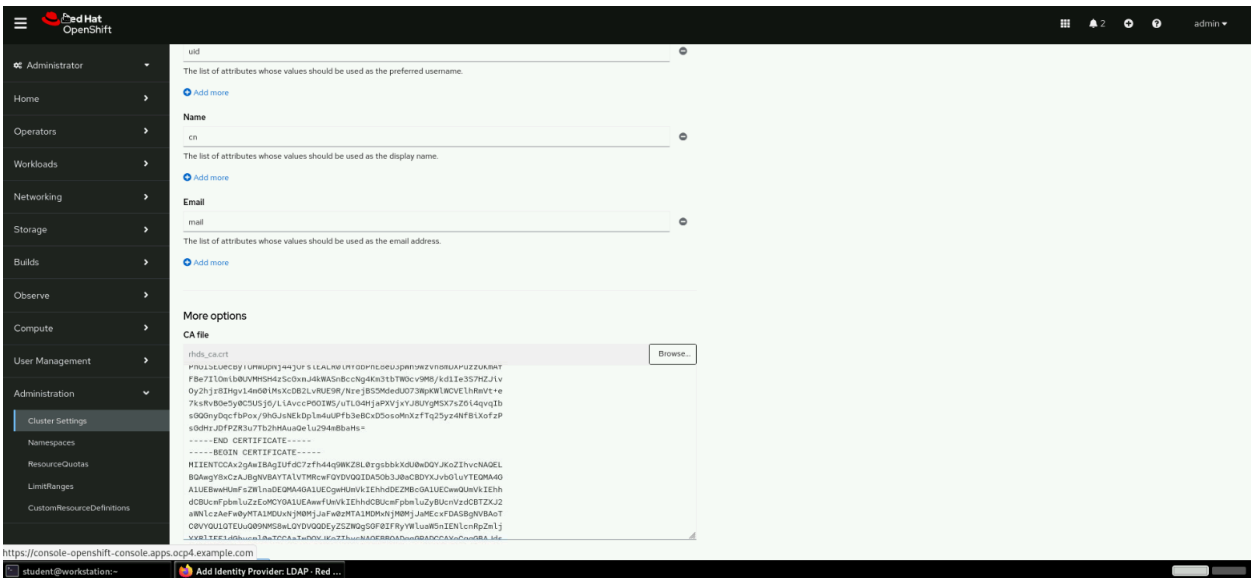
Option:2 Answer this qus by GUI Method.

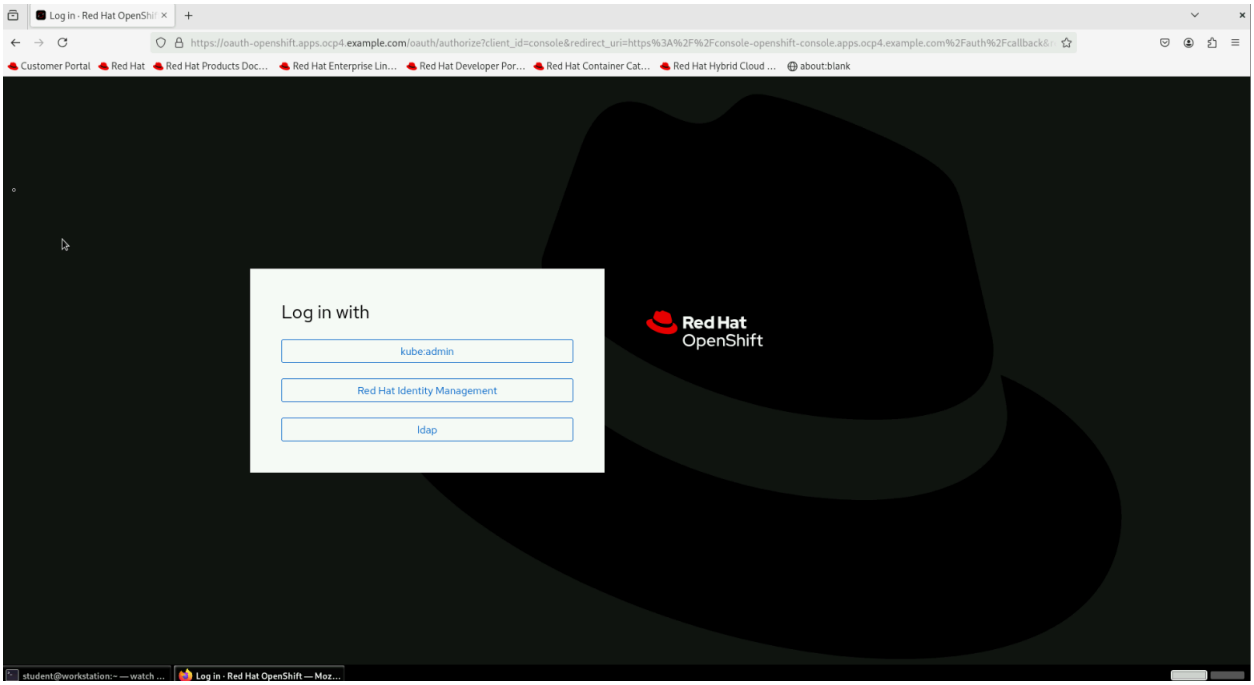
Login → Administration → Cluster Setting → Configuration → OAuth → Details → Add → LDAP → Give all info. from qus paper then → Add


Red Hat OpenShift
admin
Administrator
Home
Operators
Workloads
Networking
Storage
Builds
Observe
Compute
User Management
Administration
Cluster Settings
Namespaces
ResourceQuotas
LimitRanges
CustomResourceDefinitions
Cluster Settings
Details
ClusterOperators
Configuration
Edit the following resources to manage the configuration of your cluster.
oauth
Configuration resource
Description
Authentication
Authentication specifies cluster-wide settings for authentication (like OAuth and webhook token authenticators). The canonical name of an instance is `cluster`. Compatibility level 1: Stable within a major release for a minimum of 12 months or 3 minor releases (whichever is longer).
OAuth
OAuth holds cluster-wide information about OAuth. The canonical name is `cluster`. It is used to configure the integrated OAuth server. This configuration is only honored when the top level Authentication config has type set to IntegratedOAuth. Compatibility level 1: Stable within a major release for a minimum of 12 months or 3 minor releases (whichever is longer).


OA cluster
Details
YAML
OAuth details
Name
cluster
Labels
Edit
No labels
Basic Authentication
GitHub
GitLab
Google
HTPasswd
Keystone
LDAP
OpenID Connect
Request Header
Add
Name
Type
Mapping method
Red Hat Identity Management
LDAP
claim


Red Hat OpenShift
admin
Administrator
Home
Operators
Workloads
Networking
Storage
Builds
Observe
Compute
User Management
Administration
Cluster Settings
Namespaces
ResourceQuotas
LimitRanges
CustomResourceDefinitions
Add Identity Provider: LDAP
Integrate with an LDAP identity provider.
Name *
ldap
Unique name of the new identity provider. This cannot be changed later.
URL *
ldap://rhds.ocp4.example.com/dc=example,dc=com?uid
An RFC 2255 URL which specifies the LDAP search parameters to use.
Bind DN
cn=Directory Manager
DN to bind with during the search phase.
Bind password
••••••••••
Password to bind with during the search phase.
Attributes
Attributes map LDAP attributes to identities.
ID *
dn
The list of attributes whose values should be used as the user ID.
Add more
Preferred username
uid
The list of attributes whose values should be used as the preferred username.
Add more
Name


Red Hat OpenShift
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Networking
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Builds
Observe
Compute
User Management
Administration
Cluster Settings
Namespaces
ResourceQuotas
LimitRanges
CustomResourceDefinitions
uid
The list of attributes whose values should be used as the preferred username.
Add more
Name
cn
The list of attributes whose values should be used as the display name.
Add more
Email
mail
The list of attributes whose values should be used as the email address.
Add more
More options
CA file
rhds_ca.crt
Browse...
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
https://console-openshift-console.apps.ocp4.example.com
student@workstation:~
Add Identity Provider: LDAP - Red ...



`kristendelgado` user with `redhat123` as the password.

3. **Restore an application from a backup.**

- Restore an application from a backup The application https://lynx.apps.domain21.example.com was mistakenly deleted.
- Restore the application per the following requirements:
- The latest completed scheduled backup is used to create a restore called restore-lynx-web.
- The application https://lynx.apps.domain21.example.com is accessible using the credentials admin/Admin1andia!1958

Ans:-

Note:- For pratice this qus in lab run 2.6. Guided Exercise. `$lab start backup-restore`

```
$ oc get subs,csv,po -n openshift-adp
NAME                                                         PACKAGE
SOURCE           CHANNEL
subscription.operators.coreos.com/redhat-oadp-operator   redhat-oadp-operator
gls-catalog-cs   stable-1.4

NAME
DISPLAY            VERSION               REPLACES               PHASE
clusterserviceversion.operators.coreos.com/metallb-operator.v4.18.0-202601302
```

```
238   MetalLB Operator   4.18.0-202601302238                          Succeed
ed
clusterserviceversion.operators.coreos.com/oadp-operator.v1.4.4
OADP Operator      1.4.4                oadp-operator.v1.4.3   Succeeded

NAME                                                                 READY
STATUS      RESTARTS   AGE
pod/database-oadp-backup-1-kopia-lm99z-maintain-job-1784540291zn7z2   0/1
Completed   0          44m
pod/node-agent-6ftlc                                                 1/1
Running     0          119m
pod/node-agent-7hrsl                                                 1/1
Running     0          119m
pod/node-agent-kfchr                                                 1/1
Running     0          119m
pod/node-agent-n2njv                                                 1/1
Running     0          119m
pod/node-agent-qbj2r                                                 1/1
Running     0          119m
pod/node-agent-zjsjz                                                 1/1
Running     0          119m
pod/openshift-adp-controller-manager-9597f588-xhwlx                  1/1
Running     0          120m
pod/velero-798b8c9858-89tg4                                          1/1
Running     0          119m

$ alias velero='oc -n openshift-adp exec -it deployment/velero -c velero --
./velero'
$ velero get backup
NAME        STATUS      ERRORS   WARNINGS   CREATED                         E
XPIRES   STORAGE LOCATION   SELECTOR
db-manual   Completed   0        0          2026-07-20 08:37:36 +0000 UTC   2
9d       oadp-backup-1      <none>

$ velero get backup restore-lynx-web | grep -i namespace -A 3 lynx-web

$ vim restore.yaml
apiVersion: velero.io/v1
kind: Restore
metadata:
    name: restore-lynx-web
    namespace: openshift-adp
spec:
    backupName: db-manual
    restorePVs: true
```

```
    namespaceMapping:    # mentioned if they specified about namespace1 to na
mespace2
        lynx-web: lynx-web
EOF

$ oc apply -f restore.yaml
$ velero get restore
$ oc project
$ oc get route
```

4. Deploy and configure the GitOps Operator.

- Install and configure OpenShift GitOps to prepare for automated cluster configuration management.
  - Install the Openshift-Gitops Operator in openshift-gitops-operator namespace.
  - Use the latest stable version.
  - Configure ArgoCD Security. Configure ArgoCD server route to use reencrypt TLS Termination.
  - Setup the user access. Create a group called gitops-admin. Add cluster-admin user to this group.
  - Configure ArgoCD RBAC so that only the gitops-admins group has role:admin permission.
- Verification After completion, you should be able to access the ArgoCD web interface securely and login with administrator privileges.

Ans:-

```
$ oc whoami --show-console

Login into GUI Install gitops operator on openshift-gitops-operator namespace
using stable version.
--------------------------------------------------------------------------------
--------->

$ oc project openshift-gitops-operator
Already on project "openshift-gitops" on server "https://api.ocp4.example.co
m:6443"

$ oc get subs,csv,og,po -n openshift-gitops-operator
NAME
DISPLAY                    VERSION              REPLACES
PHASE
clusterserviceversion.operators.coreos.com/metallb-operator.v4.18.0-202601302
```

```
238   MetalLB Operator           4.18.0-202601302238
Succeeded
clusterserviceversion.operators.coreos.com/openshift-gitops-operator.v1.18.1
Red Hat OpenShift GitOps   1.18.1               openshift-gitops-operator.v
1.18.0   Succeeded

$ oc adm groups new gitops-admin
$ oc adm groups add-users gitops-admin  cluster-admin
$ oc get groups
$ oc get po -n openshift-gitops
$ oc get argocd -n openshift-gitops
NAME               AGE
openshift-gitops   5m59s
$ oc -n openshift-gitops get argocd openshift-gitops -oyaml > argocd-bkp.yaml
$ oc -n openshift-gitops edit argocds.argoproj.io
  rbac:
    defaultPolicy: ""
    policy: |
      g, system:cluster-admins, role:admin
      g, cluster-admins, role:admin
      g, gitops-admin, role:admin    # as per exam
      g, ocpadmins, role:admin       # as per lab

  server:
    autoscale:
      enabled: false
    grpc:
      ingress:
        enabled: false
    ingress:
      enabled: false
    resources:
      limits:
        cpu: 500m
        memory: 256Mi
      requests:
        cpu: 125m
        memory: 128Mi
    route:
      enabled: true
      tls:
        termination: reencrypt         # as per qus this only needs to change
    service:
      type: ""
EOF
```

```
$ oc create configmap cluster-root-ca-bundle -n openshift-gitops
  configmap/cluster-root-ca-bundle created
$ oc label configmap cluster-root-ca-bundle config.openshift.io/inject-truste
d-cabundle=true -n openshift-gitops
configmap/cluster-root-ca-bundle labeled
$ oc get cm | grep -i cluster-root-ca-bundle
$ oc edit argocd -n openshift-gitops
  repo:
    resources:
      limits:
        cpu: "1"
        memory: 1Gi
      requests:
        cpu: 250m
        memory: 256Mi
    volumeMounts:         # add from here till end
      - mountPath: /etc/pki/ca-trust/extracted/pem/tls-ca-bundle.pem
        name: cluster-root-ca-bundle
        subPath: ca-bundle.crt
    volumes:
      - configMap:
          name: cluster-root-ca-bundle
        name: cluster-root-ca-bundle
EOF

Check the repo server pod should restart.
------------------------------------------->
$ oc get po


Note: after finishing ArgoCD configuration before login, close the web-browse
r and open it again, then try login to ArgoCD.
```

5. Create and use a client certificate.

- The ACME Security company has been hired to audit your OpenShift cluster. In preparation for this audit, you need to create a client certificate that allows the auditor to examine everything on the cluster but does not allow the auditor to make any changes.
- Configure your OpenShift cluster for this audit with the blowing requirements
  - A client certificate exists with the username: **acme-auditor.**
  - A group exists with the name: **acme-corp.**

- Members of this group have access to the cluster role **cluster-reader.**
- The client certificate exists in **/home/student/kubeconfig-acme.config.**
- The client certificate must not be able to create or delete projects.
- The client certificate must be able to view all pods in the cluster.

```
This will be given in your exam.
Note:- Run the following command to generate SSL certificate.
$ openssl req -newkey rsa:4096 -nodes -keyout /home/student/acme-audit/tls.ke
y -subj "/O=acme-corp/CN=acme-auditor" -out /home/student/acme-audit/acme-aud
itor-access.csr
```

Ans:-

```
$ oc adm groups new acme-corp
group.user.openshift.io/acme-corp created

$ oc adm groups add-users acme-corp acme-auditor
group.user.openshift.io/acme-corp added: "acme-auditor"

$ openssl req -newkey rsa:4096 -nodes -keyout tls.key -subj "/O=acme-corp/CN=
acme-auditor" -out acme-auditor.csr

$ ls -lrth
tls.key    acme-auditor.csr

* To decode the certificate by base64
------------------------------------------->
$ base64 -w0 acme-auditor.csr; echo
copy the content and paste it into down file

* use below command to get kind and make in explain also.
------------------------------------------------------------->
$ oc api-resources | grep -i certificatesigning

* Craete CertificateSignerRequest file / csr.
--------------------------------------------------->
$ cat acme-auditor-csr.yaml
apiVersion: certificates.k8s.io/v1
kind: CertificateSigningRequest
metadata:
  name: acme-auditor-cert
spec:
  signerName: kubernetes.io/kubernetes-apiserver-client
  request: LS0tLS1CRUdJTiBDRVJUSUZJQ0FURSBSRVFVRVNULS0tLS0KTUlJRWNEQ0NBbGdDQV
```

```
FBd0t6RVNNQkFHQTFVRUNnd0pZV050WlMxamIzSndNUlV3RXdZRFZRUUREQXhoWTIxbApMV0YxWkd
sMGIzSXdnZ0lpTUEwR0NTcUdTSWIzRFFFQkFRVUFBNElDRHdBd2dnSUtBb0lDQVFDODJlWUJFdDgx
CldUTzJlVlZRZkRKYnVidXFuTFY4aytrSGpmR203b1RyMkpONEp6U0FZdEZPTytKNnZmL3NMWjc0b
E1ob1hvenUKaUJCMFU3dlFRZ3hOcUFEL0pBREhpcm5oczdGRmhySFAwNk8yOTVCU0dPQTh0ZlVhU3
JDZHRTTzlGNnVxckNzZgpCTkVnbTRuNGxNdTZIRld1cUNmVCt2R28xYXVNSnFGRng0UlVoU0tTS0J
tMnFjUldOWUl0WlBEUnVoa3QxMDNCCjRubU1PaVIxWUY4ejZoalVOREtWVEpyNEtHOE12SUpqazhZ
RjZVYkcxeE8zTWk0eTdBMzY3RllQMk5CeXJRdEoKUlUvUDV2Tms3ZEdYYkZCWSt6WEhjaTJEL0xRc
0RvL2hBZ2thb0F2ZUp3VzMrZUNVSzNUZ0o4b0lBYk5CRi9IMApnajRjMGx2UTEvczNqYkFxRGorZm
tGTlNrS0VqK2MrRTVIbDBGQVVGWjJUdlU1LzFId2JZTkhyWUVzbDEzSkRNClM2bndtcTlKRjZNaUo
rVW1qaFQ5WnFINjdHVG16MkRMNkdCZmQydGtPMjNTVjVEcXc1aW1uQktRTSsyaUZjYWYKMHlwN0RB
TlFEMzZ5VGZBNTJveUJSOHN3MkpXVUk3VjFkNFkxdXVJbVZhSDJBTGZLck1SalZ1QnRDQ1VOSUtDZ
ApJbXJxRXNqMWxYRDQyTENzczVnTVJoTGJGLzR6QzR5ekl0QkpNZFZNOHI3UmNoNFJLRlViV1VaY0
R3eE44NEFSCnpGWmlSNkhiLzhPWEdKdHVpTm5EaWRRQkhad3JmdDBHNUdzUmVVNWxpNUNxZHdTbnB
wUlBSVGpWRjlNdm5LeGEKM1NpdEZnVjlQdk9VcTlxYU00U2lnVXlzWTVkZ0w0TVVlUUlEQVFBQm9B
QXdEUVlKS29aSWh2Y05BUUVMQlFBRApnZ0lCQUFZRTY0WUtJY0FQbXBLZVEzZmhjazhvVWZCVUNFT
jNiSjYrWHpuWEt6WWdGRGh5YkVEQ05paTdkL01MCkErMWtxTUp5ZWI4THVGWlowT0owT2ZvM0k2Sm
JrUkVYenNGaTNGemxYc3A1Z1NPL1N4YzBlQVRzR1RSTWh2TTAKQUlrSFNIZ0U4d0RPRENwQnRVSkJ
MZUpRaGo3TFphQ2hTY3dXcE1IWks1NnZDQU44TVpQNC9uS2wyc1ZwVW0zSQovMm5TOXdjb2plN2tE
VE9FZUpDQTVid1BvbG1jME9VRXRJcjF6dXpwSDRTM1h0WVJpdEFTbzduQjNnSnlNSEJIClUwSks3a
TR4M1E0UTBCNkZGeUkzSHVaUDFKeGVmcThnWDdKUDlyNk9SRStub3lEWm9Qb3gwczJIdGRpM1pzTD
gKRVhOWnZYZG1TT1VseUJ3SWZid25TUmxyUGUrMWpvRzZWcmZQWVlSbGZkV2xpc1JENENQZ3A1aG0
4MS9hNHpxNgpaSzBRMTM5YzR6U1NnMUp0ZEdYVzBEd1F0U2VJSndBb0tQVUEvYm9PbmVQUkRzOGxw
U1VNdXQwUnBlMjFOclRnCmZqUzFuV2tTOGI3bW5pMnNBU0JkaWwwT0pZSEtCS05KT0ZGSURzWHlmN
W1sMG9BUkNKRGdXa0MwdkxGTGNYckMKZVJDM2JmYnZ5b2c1UnpXclFmaDJWTXh3OXAzbDZXZ1JPbE
1hNlZuWlNydklUcXI0cHVCaS9LQW16NWppNGF5cQpKMnFnNWwrZFQ2dURxMGJpZG1TS0wwMEE3bnZ
1am9Eay9XMXMycDdFNmk3ZlF1aFdEc0RIa3I5d2YzdkJXcEtPCkljck5MblJMdzIvMkJlMm9QQVkr
Ylh0NmtsRVZXMTVxMGF3Z1VsU0pyOUxKMDRVbAotLS0tLUVORCBDRVJUSUZJQ0FURSBSRVFVRVNUL
S0tLS0K
  expirationSeconds: 864600
  usages:
  - client auth
EOF

$ oc create -f acme-auditor-csr.yaml --dry-run=server --validate=true
$ oc create -f acme-auditor-csr.yaml

* Approve that pending CSR
------------------------------>
$ oc adm certificate approve acme-auditor-cert

* Retrive the certificate from csr decode into readble formate in openshift.
-------------------------------------------------------------------------------
--->
$ oc get csr acme-auditor-cert -o jsonpath='{.status.certificate}' | base64 -
d > acme-auditor-cert.crt
```

```
* Add the acme-auditor user credentials to the kubeconfig file.
---------------------------------------------------------------->
$ oc config set-credentials acme-auditor --client-certificate acme-auditor-ce
rt.crt --client-key tls.key --embed-certs --kubeconfig /home/student/kubeconf
ig-acme.config
User "acme-auditor" set.

* Get the public certificate from the API server.to add clusterinfo in kubeco
nfig
-------------------------------------------------------------------------------
------>
$ openssl s_client -showcerts -connect api.ocp4.example.com:6443 </dev/null 2
>/dev/null |openssl x509 -outform PEM > ocp-apiserver-cert.crt

* Set the cluster options in the kubeconfig file.
------------------------------------------------------>
$ oc config set-cluster $(oc config view  -o jsonpath='{.clusters[0].name}')
--certificate-authority ocp-apiserver-cert.crt --embed-certs=true  --server h
ttps://api.ocp4.example.com:6443 --kubeconfig /home/student/kubeconfig-acme.c
onfig
Cluster "api-ocp4-example-com:6443" set.

* Set Context for acme-auditor user
---------------------------------------->
$ oc config set-context acme-auditor --cluster $(oc config view -o jsonpath
='{.clusters[0].name}') --namespace default --user acme-auditor --kubeconfig
/home/student/kubeconfig-acme.config
Context "acme-auditor" created.

* Use the context for the acme-auditor user
----------------------------------------------->
$ oc config use-context acme-auditor --kubeconfig /home/student/kubeconfig-ac
me.config
Switched to context "acme-auditor".

$ oc whoami --kubeconfig /home/student/kubeconfig-acme.config
acme-auditor

$ oc auth can-i get pods -A --as acme-auditor --as-group acme-corp
yes

$ oc auth can-i create pods -A --as acme-auditor --as-group acme-corp
no
```

6. Configure log forwarding.

- Configure the openshift-logging project in your OpenShift cluster to forward logs to an external aggregator according to the following requirements
  - Application and infrastructure logs are forwarded to the syslog service on host `utility.lab.example.com`
  - Audit logs are forwarded to the syslog service on host `utility.lab.example.com`
  - The syslog service on both hosts listens on port `tcp/514 and port ud/514`
  - Application logs are tagged with **procid: apps**
  - Infrastructure logs are tagged with **procid: infra**
  - Audit logs are tagged with **procid: audit**
  - All logs are tagged with **appName: ocp-lab**
  - The Event Router is running using the registry image registry.domain9.example.com/openshift-logging/evenrouter-rhel9:v0.4
- The syslog servers have already been fully configured for you to receive logging messages and filter them using the procID attributes to following files
  /var/log/ex380-app.log
  /var/log/ex380-intra.log
  /var/log/ex380-audit.log
  You may check your work by logging into the syslog servers using the opsadm user and admin password. You dont have root access to this machines.

Ans:-

```
$ lab start logging-forward  (IN LAB)

Install the Logging Operator in openshift-logging namespace.
-------------------------------------------------------------->

$ oc -n openshift-logging get subs,csv,po
NAME                                                PACKAGE           SOURCE
CHANNEL
subscription.operators.coreos.com/cluster-logging   cluster-logging   gls-cat
alog-cs   stable-6.4

NAME
DISPLAY                     VERSION              REPLACES   PHASE
clusterserviceversion.operators.coreos.com/cluster-logging.v6.4.0
Red Hat OpenShift Logging   6.4.0                           Succeeded
clusterserviceversion.operators.coreos.com/metallb-operator.v4.18.0-202601302
238   MetalLB Operator            4.18.0-202601302238              Succeeded

NAME                                          READY   STATUS    RESTARTS
AGE
```

```
pod/cluster-logging-operator-78bc55f6cc-4j4ht   1/1     Running   0
63s


$ oc create sa log-collector
serviceaccount/log-collector created
$ oc adm policy add-cluster-role-to-user collect-application-logs -z log-coll
ector
$ oc adm policy add-cluster-role-to-user collect-audit-logs -z log-collector
$ oc adm policy add-cluster-role-to-user collect-infrastructure-logs -z log-c
ollector

Create ClusterLogForwarder (clf) from GUI copy the yaml and apply it.
$ vim clf.yaml
apiVersion: observability.openshift.io/v1
kind: ClusterLogForwarder
metadata:
  name: instance
  namespace: openshift-logging
spec:
  managementState: Managed
  outputs:
    - syslog:
        rfc: RFC5424
        url: 'tcp://utility.lab.example.com:514'
        appName: ocp-lab
        msgId: apps   # in exam instead of msgid it procID
      name: app-syslog-tcp
      type: syslog
    - syslog:
        rfc: RFC5424
        url: 'udp://utility.lab.example.com:514'
        appName: ocp-lab
        msgId: apps  #in exam instead of msgid it procID
      name: app-syslog-udp
      type: syslog

    - syslog:
        rfc: RFC5424
        url: 'tcp://utility.lab.example.com:514'
        appName: ocp-lab
        msgId: infra  # in exam instead of msgid it procID
      name: infra-syslog-tcp
      type: syslog
    - syslog:
```

```
        rfc: RFC5424
        url: 'udp://utility.lab.example.com:514'
        appName: ocp-lab
        msgId: infra       # in exam instead of msgid it procID
      name: infra-syslog-udp
      type: syslog

    - syslog:
        rfc: RFC5424
        url: 'tcp://utility.lab.example.com:514'
        appName: ocp-lab
        msgId: audit # in exam instead of msgid it procID
      name: audit-syslog-tcp
      type: syslog
    - syslog:
        rfc: RFC5424
        url: 'udp://utility.lab.example.com:514'
        appName: ocp-lab
        msgId: audit
      name: audit-syslog-udp
      type: syslog
  pipelines:
    - inputRefs:
        - application
      outputRefs:
        - app-syslog-tcp
        - app-syslog-udp
      name: app-syslog
    - inputRefs:
        - infrastructure
      outputRefs:
        - infra-syslog-tcp
        - infra-syslog-udp
      name: infra-syslog
    - inputRefs:
        - audit
      outputRefs:
        - audit-syslog-tcp
        - audit-syslog-udp
      name: audit-syslog
  serviceAccount:
    name: log-collector
EOF

$ oc get daemonsets.apps
```

```
NAME       DESIRED   CURRENT   READY   UP-TO-DATE   AVAILABLE   NODE SELECTOR
AGE
instance   6         6         6       6            6           kubernetes.i
o/os=linux   12m
$ oc get po
NAME                                          READY   STATUS    RESTARTS   AGE
cluster-logging-operator-78bc55f6cc-4j4ht     1/1     Running   0          60m
instance-5gs8q                                1/1     Running   0          6m29
s
instance-8227q                                1/1     Running   0          6m29
s
instance-dj2rh                                1/1     Running   0          6m29
s
instance-qhqgp                                1/1     Running   0          6m29
s
instance-sq9td                                1/1     Running   0          6m29
s
instance-vkp2h                                1/1     Running   0          6m29
s



In Exam there will be documentation (which is for all), You need to find even
trouter template file form there and then in that template file chnage your e
ventrouter image name then apply it.

$ vim eventrouter.yaml
$ oc process -f eventrouter.yaml | oc apply -f -
$ oc get po

You can login into syslog host and validate your file has been created.
```

7. Create an ArgoCD managed application.

- Configure Git Repository in ArgoCD
  - Add the Git repository `https://git.ocp4.example.com/developer/etherpad-app.git` to ArgoCD
  - Important Enable Skip server verification for this repository
  - The MachineConfig manifests are stored in this repository
- **Create ArgoCD Application**
  - Create an ArgoCD application named machineconfig-motd-deploy
  - Configure it to use the sshd-motd path in the Git repository
  - Critical: Set the Sync Policy to Manual (not automatic)
  - Target the default cluster (in-cluster deployment)

- **Create a MachineConfig for Master Nodes**
  - Create a file named 71-master-sshd-motd.yaml in the Git repository with these specifications
  - Target Configure this MachineConfig to apply only to master nodes using label **machineconfiguration.openshift.io/role: master**
  - Ignition: Use the latest ignition version available on the cluster
  - File Creation Configure the MachineConfig to create /etc/motd with these properties
  - Set file permissions to r-r-r-
  - Set overwrite existing file to: true
  - Content source: Configure content from: **http://rhgls.domain9.example.com/ssh.motd**
- **Create a MachineConfig for Worker Nodes**
  - Create a file named 71-worker-sshd-motd.yaml in the Git repository with these specifications
  - Target Configure this MachineConfig to apply only to worker nodes using label **machineconfiguration.openshift.io/role: worker**
  - Configuration: Use identical settings as the master MachineConfig (same ignition version file properties and content source
  - Set file permissions to: r-r-r-
  - Set overwrite existing file to: true
  - Content source: Configure content from **http://rhgls.domain9.example.com/ssh.motd**
- Deploy and Verify
  - Commit and push both MachineConfig files to the Git Repository
  - Update the kustomization.yaml file to include both new MachineConfig files
  - Manually sync the ArgoCD application to deploy the configurations
  - Verify that you can find the /etc/motd file on all cluster nodes with correct permissions.

Ans:-

1. For lab configuration according to exam. Run `lab start gitops-app`
2. Login into Git repository and create a public project **ocp-gitops** using the credential **project-admin** and password is **r3dh4tgit.**
3. In exam you need to download the file from given url called **ssh.motd** but in our lab we need to created it manually.

```
$ mkdir git;cd git
$ wget http://rhgls.domain9.example.com/ssh.motd
------------------------------------------------------------->
for lab this ssh.motd file we created manually using below cmd.
$ echo "Welcome to Cluster | OpenShift" >> ssh.motd
------------------------------------------------------------->
```

```
$ ls
ssh.motd

Generate encoded file content using below cmd
------------------------------------------------------->
$ $ cat ssh.motd | base64 -w0 ; echo
V2VsY29tZSB0byBDbHVzdGVyIHwgT3BlblNoaWZ0Cg==

$ vim 71-master-sshd-motd.yaml
apiVersion: machineconfiguration.openshift.io/v1
kind: MachineConfig
metadata:
  labels:
    machineconfiguration.openshift.io/role: master
  name: 71-master-sshd-motd
spec:
  config:
    ignition:
      version: 3.4.0
    storage:
      files:
      - contents:
          source: data:text/plain;charset=utf-8; base64,V2VsY29tZSB0byBDbHVzdG
VyIHwgT3BlblNoaWZ0Cg==
        mode: 0444
        overwrite: true
        path: /etc/motd
EOF

$ vim 71-worker-sshd-motd.yaml
apiVersion: machineconfiguration.openshift.io/v1
kind: MachineConfig
metadata:
  labels:
    machineconfiguration.openshift.io/role: worker
  name: 71-master-worker-motd
spec:
  config:
    ignition:
      version: 3.4.0
    storage:
      files:
      - contents:
          source: data:text/plain;charset=utf-8; base64,V2VsY29tZSB0byBDbHVzdG
VyIHwgT3BlblNoaWZ0Cg==
```

```
        mode: 0444
        overwrite: true
        path: /etc/motd
EOF

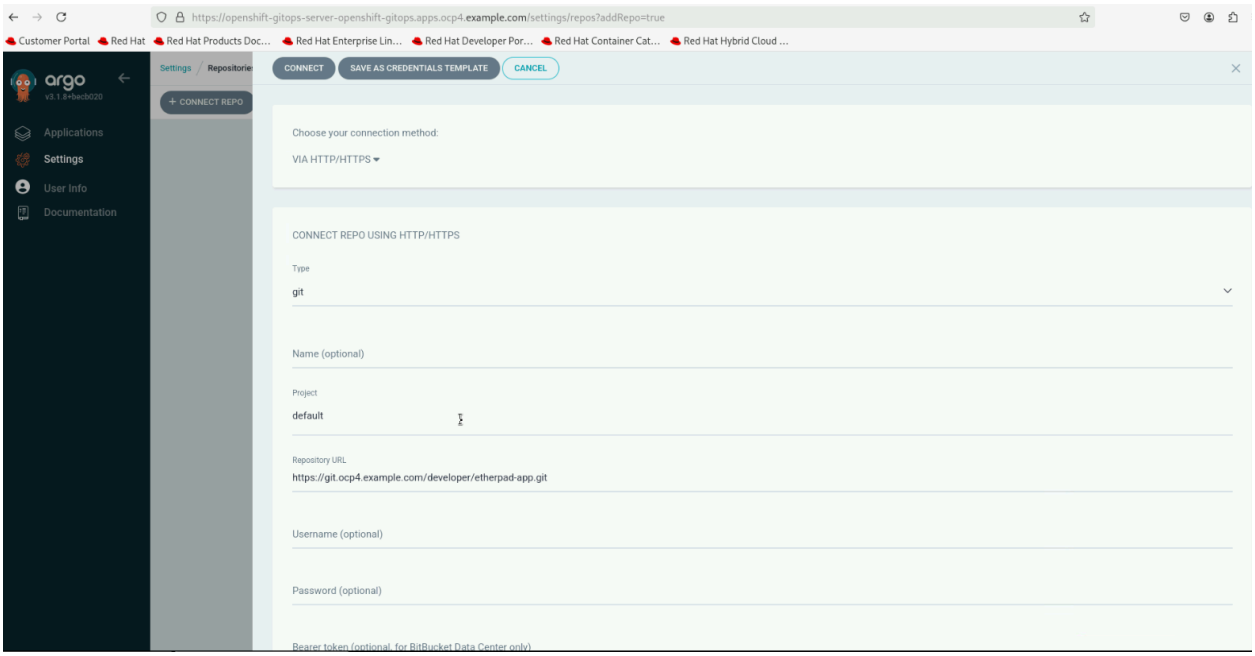
$ git clone https://git.ocp4.example.com/developer/etherpad-app.git

Note-1:
After cloning the Git repository, you will find a directory containing anothe
r subdirectory. Navigate to the subdirectory and place both MachineConfig YAM
L files there.

Note-2:
In the same subdirectory, you will find a kustomization.yaml file. Edit this
file and add the names of both MachineConfig YAML files under the resources s
ection.

$ cp 71* ocp-gitops
$ cd ocp-gitops
$ git add 71-worker-sshd-motd.yaml 71-master-sshd-motd.yaml kustomization.yam
l
$ git commit -m 'sshd-motd'
$ git push

Now login into Argocd. 1st add the repository
ArgoCD -> Login -> Setting -> Repository -> connect repo -> add from below sc
reenshoft -> connect
```



Now Create application.

ArgoCD → Login → New App → Fill the info → Create